

Vidura Abeysinghe

Software Engineering Undergraduate

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EDUCATION

Bachelor of Software Engineering (Hons)

Expected Graduation - June 2027

The Open University of Sri Lanka

Relevant Coursework: Software Development & Engineering, Data Structures & Algorithms, Data Modelling & Database Systems, Artificial Intelligence, Data Science, Performance Modelling, Agent Technology, Cloud Computing, Information Security

Chartered Accountancy - Level 2

2023

Institute of Chartered Accountants of Sri Lanka

WORK EXPERIENCE

SRI LANKA TELECOM

Colombo, Sri Lanka

AI & ML Engineer - Internship

Sep 2025 - Apr 2026

- Engineered conversational AI agents utilizing LangGraph and the OpenAI API, implementing semantic caching to optimize LLM inference latency and reduce redundant API calls.
- Secured LLM pipelines utilizing Guardrails AI to deploy regex matching, SQL validation, and state-aware sequence monitoring, preventing hallucinations and sequence violations.
- Containerized database systems (PostgreSQL, Neo4j, ChromaDB) and FastAPI backend using Docker, developing automated Python data pipelines to support enterprise knowledge retrieval and RAG architectures.

PROJECTS & CONTRIBUTIONS

Enterprise AI Agent

- Developed an AI powered enterprise assistant for knowledge retrieval and product intelligence using graph and vector database populated using web scrapers.
 - Implemented conversational AI workflows with LangGraph and OpenAI API, integrating guardrails for secure interactions.
- Technologies: Python, LangGraph, Chroma DB, Neo4j, OpenAI, React, PostgreSQL, Docker, Guardrails

Fuel Station Management System

- Developed an enterprise fuel station management system to digitize daily operations, sales information and inventory tracking.
- Firebase Authentication and role-based authorization.

Technologies: Next.js, React.js, TypeScript, Tailwind CSS, Firebase (Firestore & Auth)

OpenSesame Academic Platform (Live Demo)

- A web-based platform designed to help with the academic experience of students at the Open University of Sri Lanka (OUSL).
- Technologies: JavaScript, HTML, Cascading Style Sheets (CSS)

CashCoach - Mobile app

- Developed an Android application to help users track and manage daily expenses.

Technologies: Java, Android Studio, Firebase

Agentic GraphRAG Study Partner

- Adaptive GraphRAG-based study platform designed to identify student knowledge gaps and personalize exam preparation.
- Combines probabilistic mastery tracking with multilingual terminology support to deliver explanations.

Technologies: Python, LangGraph, GraphRAG, Neo4j, ChromaDB, PostgreSQL, FastAPI, React.js, spaCy, SimPy

ADDITIONAL

• **Technical Skills:** Python, C#, Java, JavaScript, React.js, TypeScript, Tailwind CSS, LangChain, LangGraph, ASP.NET Core, Node.js, Express.js, FastAPI, MySQL, MongoDB, PostgreSQL, Firebase, ChromaDB, Neo4j, SQLite, Pandas, NumPy, Matplotlib, Selenium, GitHub, Docker, Postman, Jira, Figma

• **Publications:** Green Mart: A user-centered mobile application for sustainable online shopping

• **Certifications:** Quality Assurance - Track (SLASSCOM), Introduction to Machine Learning, Data Science & Analytics